

Intermediate Algebra Practice Midterm

December 5, 2023

Problem 1

Find each product or quotient:

1. $(-3)(-6)$

2. $(-12)(-2)$

3. $(3)(-11)$

4. $\frac{-56}{-8}$

5. $\frac{-32}{2}$

Problem 2

Evaluate each expression.

1. $\frac{2}{3} + (-\frac{8}{3})$

2. $\frac{3}{5} + \frac{5}{3}$

3. $\frac{11}{8} - \frac{5}{2}$

4. $\frac{9}{5} - (-\frac{5}{3})$

5. $(-2)(-\frac{7}{6})$

Problem 3

Evaluate each expression.

1. $\frac{-9-(2-5)}{-6}$

2. $[-1 - (-9)]|3 + 2|$

3. $(-7 - 9) \div [-2 - 2 - (-8)]$

4. $3(-5 + 6) \cdot 5^2$

5. $\frac{-13-2}{2-(-1)^3+(-6)-[-1-(-3)]}$

Problem 4

Evaluate: $p + (q - r)(6 - p)$ where $p = 4$, $q = 5$, and $r = 5$.

Problem 5

Solve: $\frac{r}{13} = \frac{12}{39}$.

Problem 6

Solve: $\frac{b}{5} + 4 = 10$.

Problem 7

Solve: $-7(x - 2) = -4 - 6(x - 1)$.

Problem 8

Solve: $-a - \frac{5}{4}\left(-\frac{8}{3}a + 1\right) = -\frac{19}{4}$.

Problem 9

Solve: $ax + bx = c$ for b .

Problem 10

Solve: $|5x + 1| = |2x - 15|$

Problem 11

Find the constant of variation and write the formula to express the relationship using that constant given that: t varies directly as the square of u , and $t = 6$ when $u = 3$.

Problem 12

A number plus itself, plus twice itself, plus 4 times itself, is equal to -112. What is the number?

Problem 13

A is now 34 years old, and B is 14 years old. In how many years will A be twice as old as B ?

Problem 14

A is 60 miles from B . An automobile at A starts for B at the rate of 20 miles an hour at the same time that an automobile at B starts for A at the rate of 25 miles an hour. How long will it be before the automobiles meet?

Problem 15

Sketch the graph of $y = -\frac{5}{4}x - 4$. CLEARLY label at least two points on the graph.

Problem 16

Find the slope of the line through the points $(-2, 10)$ and $(-2, -15)$.

Problem 17

Write the slope-intercept form of the equation $-3y = -5x + 9$.

Problem 18

Write the point-slope form of the equation of the line through the point $(-1, -5)$ with a slope of 9.

Problem 19

Write the slope-intercept form of the equation of the line through the point $(-2, 5)$, and perpendicular to the line $2x + y = 3$.

Problem 20

Solve the compound inequality, graph the solution, and give interval notation;

$$\frac{u}{3} > -2 \text{ and } u - 3 < 2.$$

Problem 21

Solve the inequality, graph the solution, and give interval notation;

$$\frac{m}{5} \leq -\frac{6}{5}.$$

Problem 22

An executive drove from home at an average speed of 40 mph to an airport where a helicopter was waiting. The executive boarded the helicopter and flew to the corporate offices at an average speed of 60 mph. The entire distance was 150 mi. The entire trip took 3 hrs. Find the distance from the airport to the corporate offices.

Problem 23

Solve the inequality, graph the solution, and give interval notation;

$$3 - 2|x - 5| \leq -15$$

Problem 24

Solve the system of equations;

$$-3x + 3y = -12$$

$$-3x + 9y = 24.$$

Problem 25

Solve the system of equations;

$$-10x - 5y = 0$$

$$-x + 4y = -17.$$

Problem 26

Solve the system of equations;

$$3x + y - z = 11$$

$$x + 3y = z + 13$$

$$x + y - 3z = 11.$$

Problem 27

A collection of dimes and quarters is worth \$15.25. There are 103 coins in all. How many of each is there?

Problem 28

A tank contains 8000 liters of a solution that is 40% acid. How much water should be added to make a solution that is 30% acid?

Problem 29

Simplify:

$$\frac{2x^7y^5}{3x^3y \cdot 4x^2y^3}.$$

Problem 30

Simplify:

$$\frac{y^3 \cdot x^{-3}y^2}{(x^4y^2)^3}.$$

Problem 31

Simplify in scientific notation:

$$\frac{5.33 \times 10^{-6}}{9.62 \times 10^{-2}}.$$

Problem 32

Simplify: $(5n^4 + 6n^3) + (8 - 3n^3 - 5n^4)$.

Problem 33

Simplify: $(6n - 4)(2n^2 - 2n + 5)$.

Problem 34

Simplify: $(4v - 7)(4v + 7)$.

Problem 35

Divide:

$$\frac{3n^3 + 9n^2 - 64n - 68}{n + 6}.$$

Problem 36

Simplify: $(2a^4)^4$.

Problem 37

Divide:

$$\frac{x^3 - 16x^2 + 71x - 56}{x - 8}.$$

Problem 38

There were 200 tickets sold for a women's basketball game. Tickets for students were \$0.50 each and for adults \$0.75 each. The total amount of money collected was \$132.50. How many of each type of ticket was sold?

Problem 39

Solve the inequality, graph the solution, and give interval notation; $6 - 3|1 - 4x| < -3$.

Problem 40

Solve the system of equations;

$$w + x + y + z = 2$$

$$w + 2x + 2y + 4z = 1$$

$$-w + x - y - z = -6$$

$$-w + 3x + y - z = -2.$$